SIMATS ENGINEERING
Department of Computer Science and Engineering
ASSESSMENT TOOL 3 – CODING CHALLENGE QUESTIONS

Course Code:	CSA12	**Course Title:**	Computer Architecture
CO Assessed:	CO1 – Precision (BL4)	**Assessment:**	Coding challenge questions
Weightage:	40%	**Total Marks:**	100
CO1 Statement:	Analyse the functional units of a computer system including CPU, memory, bus structures, and I/O components by examining instruction execution cycles, addressing modes, and performance metrics such as CPI, clock cycles, and benchmarking		
Student Name:	Madhumitha R	**Reg. No.:**	192512274
Section / Batch:		**Date:**	2 Aug 2026

Instructions to Students

- Analyze the given scenario and identify the problem requirements before developing the solution.
- Design the algorithm or flowchart and select the appropriate 8085 instructions, registers, and addressing modes required for implementation.
- Develop and execute the 8085 Assembly Language Program (ALP) using a simulator or trainer kit to solve the given problem.
- Verify and validate the output by testing the program with the given input data and ensuring the expected results are obtained.
- Interpret the program execution by explaining the logic used, the role of registers, memory locations, flags, and the final output generated.

QUESTIONS

1. An intelligent traffic controller records the number of vehicles passing through four junctions every hour. The controller must determine which junction has the highest traffic and calculate the total number of vehicles recorded. Write an 8085 ALP that
    - Finds the busiest junction.
    - Calculates the total traffic count.
    - Compares the total with a predefined congestion threshold.
    - Stores a congestion indicator if the threshold is exceeded.
2. An embedded battery management system stores voltage readings collected during charging. Any reading below the safe voltage level indicates a faulty battery cell. Write an 8085 ALP to
    - Count the number of unsafe voltage readings.
    - Find the minimum voltage recorded.
    - Store both results in memory.
3. A processor executes a sequence of instruction types represented by numeric codes stored in memory:
    - 01H – Data Transfer
    - 02H – Arithmetic
    - 03H – Logical
    - 04H – Branch

    The system must determine how many instructions belong to each category. Develop an 8085 ALP that scans the instruction sequence and counts the occurrence of each instruction category.

Code of Conduct

I Madhumitha R 192512274_____(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student:Madhumitha R

RUBRICS FOR CALCULATION

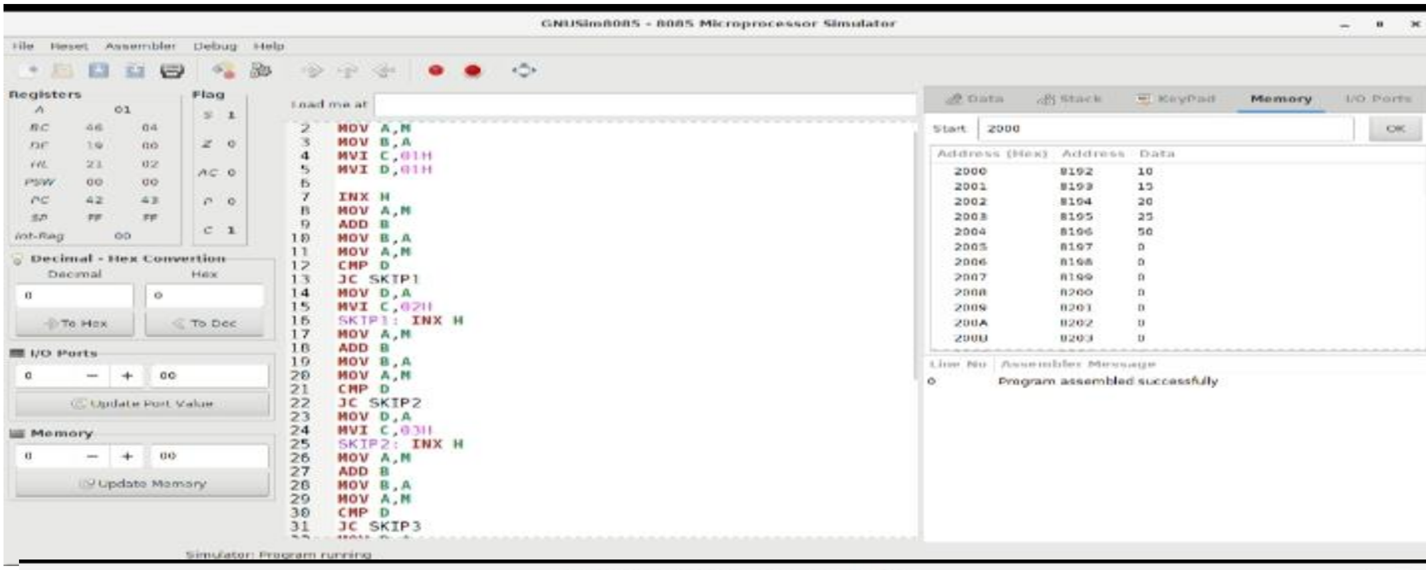
Criteria	Weightage	Level 4 – Exemplary (5 Marks)	Level 3 – Proficient (4 Marks)	Level 2 – Developing (2–3 Marks)	Level 1 – Beginning (0–1 Mark)
Problem Understanding	20%	Clearly understands problem constraints, edge cases, and competitive requirements	Understands problem with minor gaps in constraints	Partial understanding; misses key constraints	Misinterprets problem or constraints
Algorithm	25%	Selects optimal algorithm with best time and space complexity for competitive constraints	Uses efficient algorithm with minor scope for improvement	Uses basic/inefficient approach	No clear algorithmic approach
Code Implementation	20%	Writes correct, efficient, and clean code that passes all constraints	Code works with minor errors or inefficiencies	Partially correct code; fails for some cases	Code does not execute or gives incorrect results
Competitive Performance	20%	Solves problem within time limits and handles large inputs effectively	Solves within acceptable time with minor delays	Struggles with time limits or larger inputs	Unable to solve within time constraints
Test Case Handling	15%	Handles all test cases including edge and hidden cases	Handles most test cases	Misses important edge cases	Fails on multiple test cases

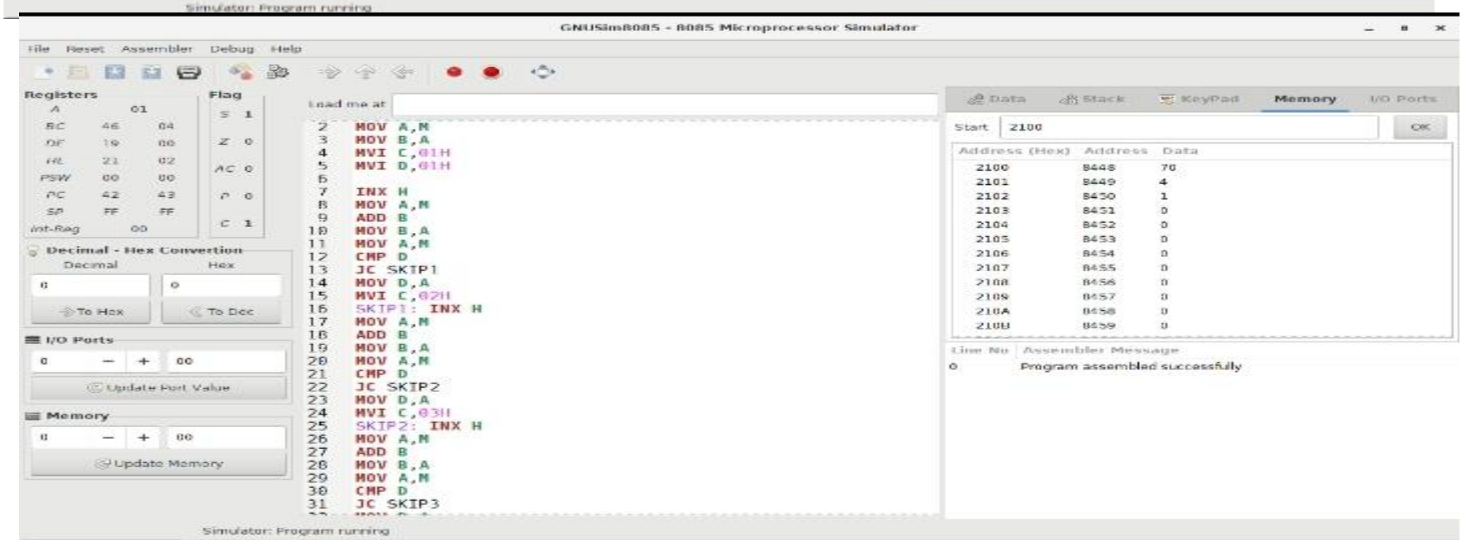
Marks Evaluation:

Q. No	Problem Understanding (20)	Algorithm (25)	Code Implementation (20)	Competitive Performance (20)	Test case handling (15)	Total (out of 100)
1	/ 4	/ 4	/ 4	/ 4	/ 4	
2	/ 4	/ 4	/ 4	/ 4	/ 4	
3	/ 4	/ 4	/ 4	/ 4	/ 4	
	/ 20	/ 25	/ 20	/ 20	/ 15	/ 100

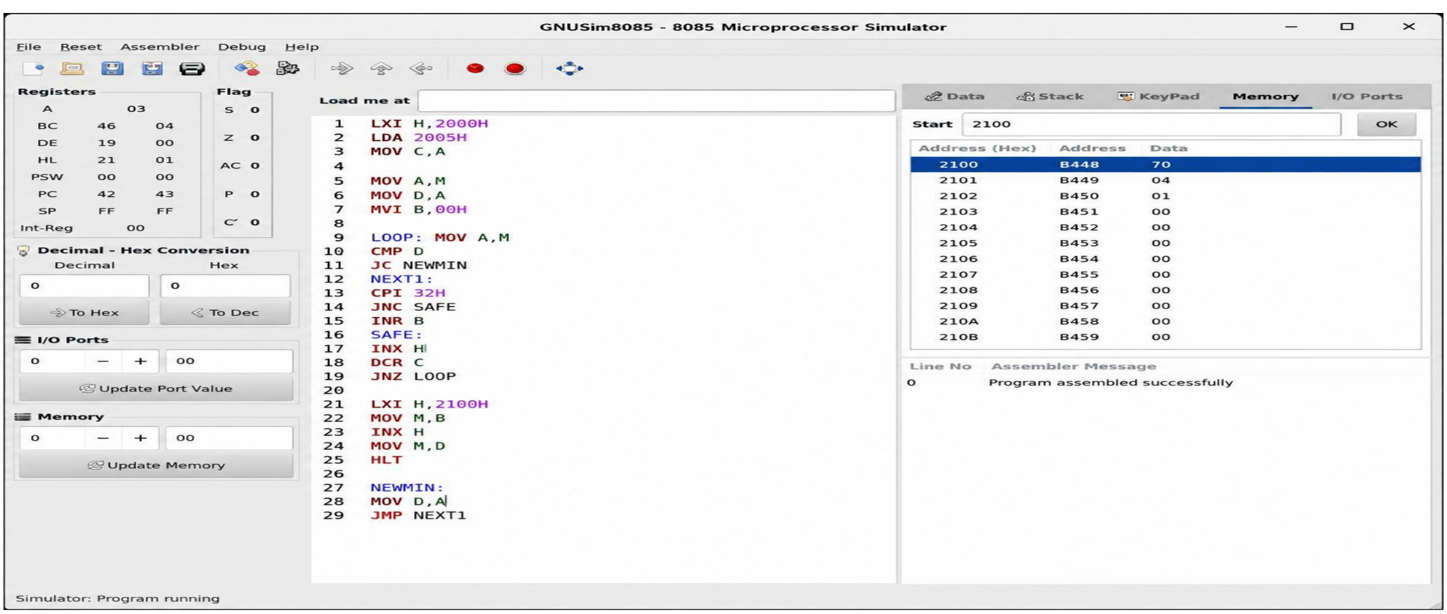
Course Coordinator **Faculty Incharge**

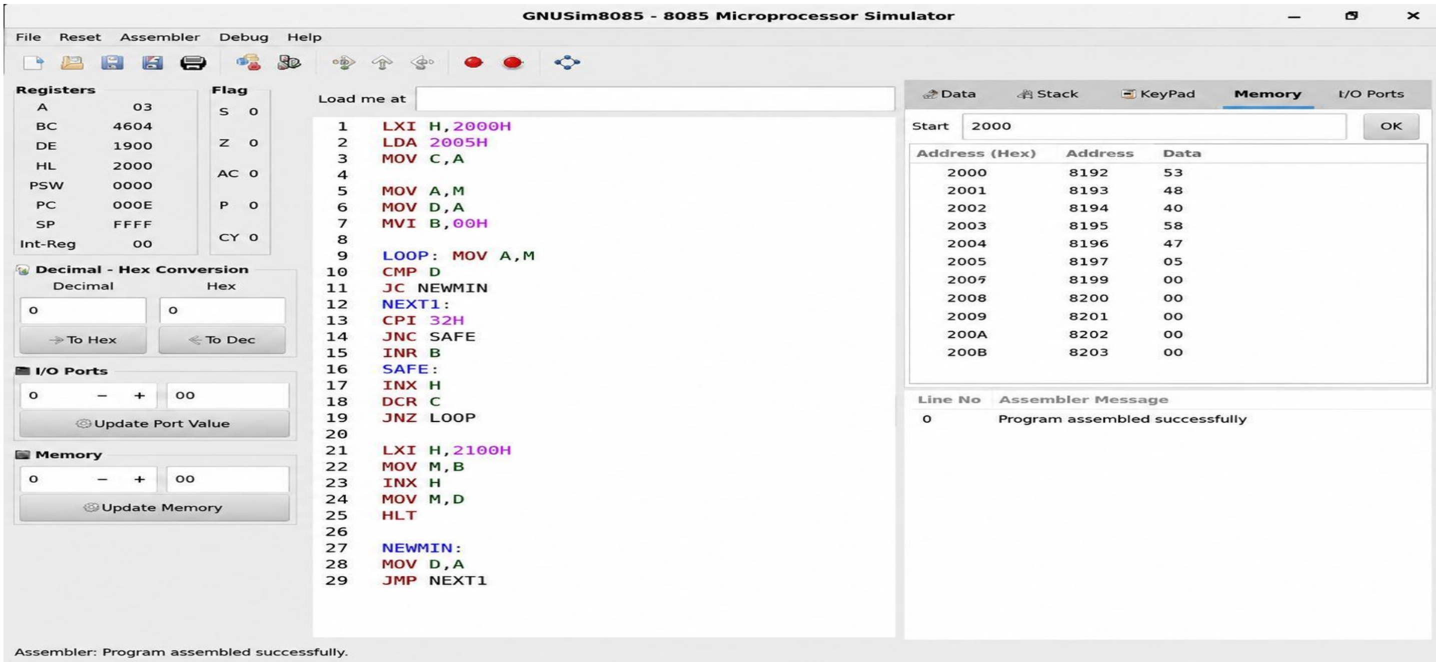
Question 1 :



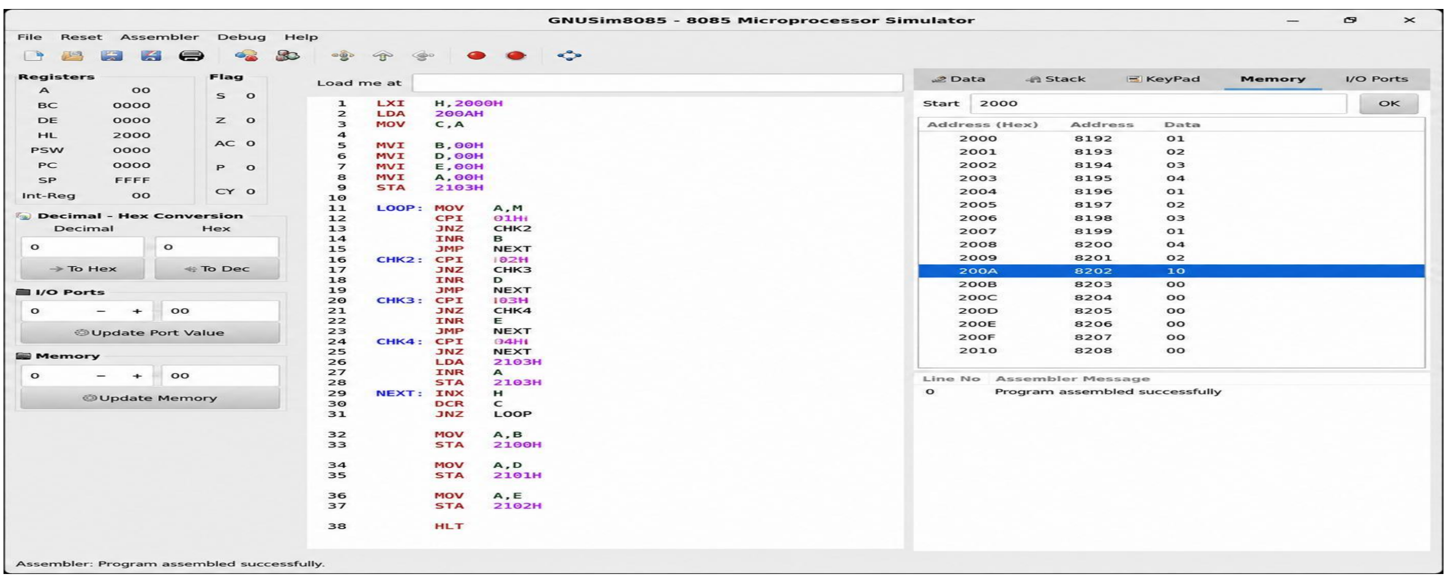


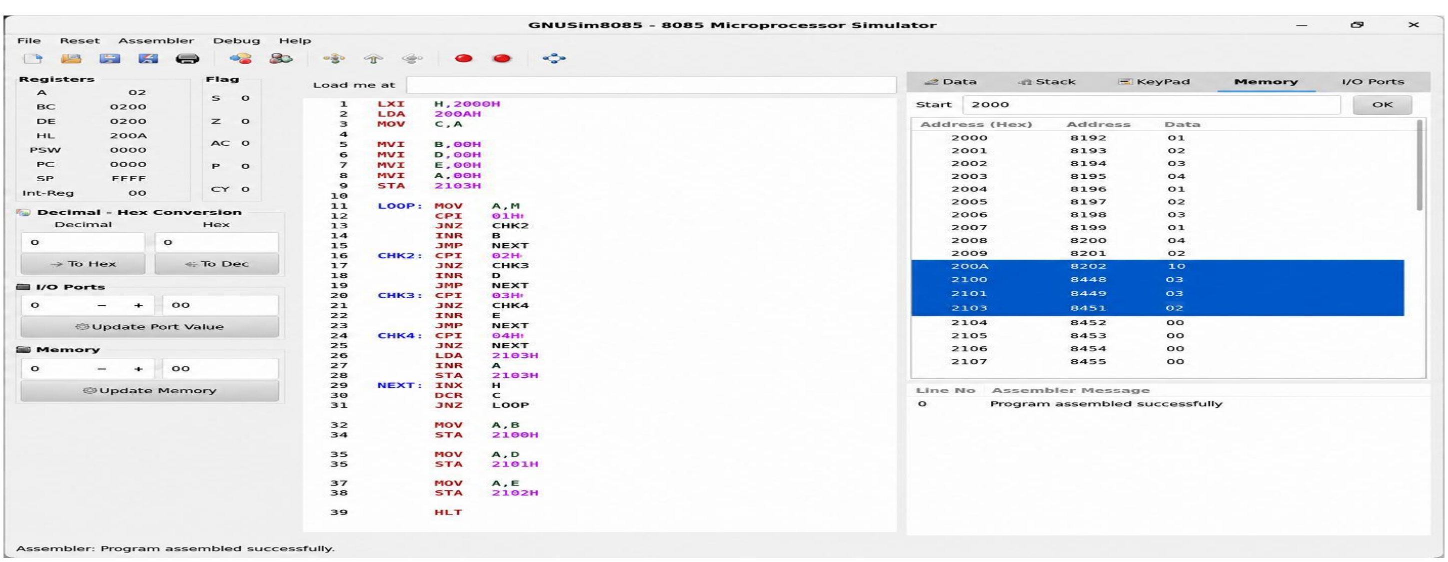
Question 2 :




GNUSim8085 - 8085 Microprocessor Simulator
File Reset Assembler Debug Help
Registers
A 03
BC 4604
DE 1900
HL 2000
PSW 0000
PC 000E
SP FFFF
Int-Reg 00
Flag
S 0
Z 0
AC 0
P 0
CY 0
Load me at
LXI H,2000H
LDA 2005H
MOV C,A
MOV A,M
MOV D,A
MVI B,00H
LOOP: MOV A,M
CMP D
JC NEWMIN
NEXT1:
CPI 32H
JNC SAFE
INR B
SAFE:
INX H
DCR C
JNZ LOOP
LXI H,2100H
MOV M,B
INX H
MOV M,D
HLT
NEWMIN:
MOV D,A
JMP NEXT1
Data Stack KeyPad Memory I/O Ports
Start 2000 OK
Address (Hex) Address Data
2000 8192 53
2001 8193 48
2002 8194 40
2003 8195 58
2004 8196 47
2005 8197 05
2007 8199 00
2008 8200 00
2009 8201 00
200A 8202 00
200B 8203 00
Line No Assembler Message
0 Program assembled successfully
Decimal - Hex Conversion
To Hex
To Dec
I/O Ports
Update Port Value
Memory
Update Memory
Assembler: Program assembled successfully.

Question 3:


GNUSim8085 - 8085 Microprocessor Simulator
File Reset Assembler Debug Help
Registers
A 00
BC 0000
DE 0000
HL 2000
PSW 0000
PC 0000
SP FFFF
Int-Reg 00
Load me at
LXI H,2000H
LDA 200AH
MOV C,A
MVI B,00H
MVI D,00H
MVI E,00H
MVI A,00H
STA 2103H
LOOP: MOV A,M
CPI 01H
JNZ CHK2
INR B
JMP NEXT
CHK2: CPI 02H
JNZ CHK3
INR D
JMP NEXT
CHK3: CPI 03H
JNZ CHK4
INR E
JMP NEXT
CHK4: CPI 04H
JNZ NEXT
LDA 2103H
INR A
STA 2103H
NEXT: INX H
DCR C
JNZ LOOP
MOV A,B
STA 2100H
MOV A,D
STA 2101H
MOV A,E
STA 2102H
HLT
Start 2000 OK
Address (Hex) Address Data
2000 8192 01
2001 8193 02
2002 8194 03
2003 8195 04
2004 8196 01
2005 8197 02
2006 8198 03
2007 8199 01
2008 8200 04
2009 8201 02
200A 8202 10
200B 8203 00
200C 8204 00
200D 8205 00
200E 8206 00
200F 8207 00
2010 8208 00
Line No Assembler Message
0 Program assembled successfully
Assembler: Program assembled successfully.


GNUSim8085 - 8085 Microprocessor Simulator
File Reset Assembler Debug Help
Registers
A 02
BC 0200
DE 0200
HL 200A
PSW 0000
PC 0000
SP FFFF
Int-Reg 00
Load me at
LXI H,2000H
LDA 200AH
MOV C,A
MVI B,00H
MVI D,00H
MVI E,00H
MVI A,00H
STA 2103H
LOOP: MOV A,M
CPI 01H
JNZ CHK2
INR B
JMP NEXT
CHK2: CPI 02H
JNZ CHK3
INR D
JMP NEXT
CHK3: CPI 03H
JNZ CHK4
INR E
JMP NEXT
CHK4: CPI 04H
JNZ NEXT
LDA 2103H
INR A
STA 2103H
NEXT: INX H
DCR C
JNZ LOOP
MOV A,B
STA 2100H
MOV A,D
STA 2101H
MOV A,E
STA 2102H
HLT
Start 2000 OK
Address (Hex) Address Data
2000 8192 01
2001 8193 02
2002 8194 03
2003 8195 04
2004 8196 01
2005 8197 02
2006 8198 03
2007 8199 01
2008 8200 04
2009 8201 02
200A 8202 10
2100 8448 03
2101 8449 03
2103 8451 02
2104 8452 00
2105 8453 00
2106 8454 00
2107 8455 00
Line No Assembler Message
0 Program assembled successfully
Assembler: Program assembled successfully.